

WRM Analysis: The Boötes Void as a Wave Node

In the Wave Refraction Model (WRM), a Boötes void is defined not as a mere defect in matter, but as a **"node of deterministic interference fringes" created by cosmic-scale external waves.**

1. Geometric correlation with the dipole axis

The Bootis cavity is located at an angle of approximately 70–80 degrees from the CMB dipole axis (the principal gradient axis of the universe). This coincides with the theoretical threshold in wave geometry where the first destructive interference occurs from the central axis.

2. Derivation of cavity size using $\alpha = 0.046$

The cavity diameter L_{void} matches with astonishing accuracy the "half-wavelength" calculation result using the characteristic length of the universe (Hubble radius $R_H \approx 14.0$ Gly) and the correction term α .

$$L_{\text{void}} \approx (R_H \times \alpha) / 2$$

- Calculated value:** $(14 \text{ billion light-years} \times 0.046) / 2 = 322 \text{ million light-years}$
- Observed values:** Estimated diameter of Bootis Cavity $\approx 330 \text{ million light-years}$

This coincidence physically proves that the cavity is the "node-to-node interval" where the energy of the external wave is minimized.

3. Mechanism of substance exclusion

In the cavity area, **canceled interference** occurs where external waves overlap at 180 degrees opposite phase. This flattens the refractive index gradient of space (i.e., the gravitational potential), and material is pushed towards the filament (constructive interference region) where the wave pressure is stronger. As a result, a geometric void defined by α is formed.

project	Standard cosmology (bubble structure)	WRM (Wave Refraction Model)
Causes	Random accumulation due to gravity	Cancellation interference of external waves (deterministic)
Basis for size	Statistical probability	$R_H \times \alpha / 2$ (Physical constant dependency)
Meaning of a cavity	Material deficiency	The "nodes" of the wave potential

Direct Translation: The Boötes Void is a deterministic "node" of the external wave in WRM. Its size (3.3 Gly) matches the calculation of half the product of the Hubble radius and $\alpha=0.046$, proving it is a physical result of destructive interference.

WRM Structural Scaling Analysis

Analysis of the 60 isolated galaxies in the Boötes Void via $\alpha=0.046$

1. Derivation of Scaling Law from Determinism

The distribution scale of the "60 galaxies (galaxy group)" located within the Boötes void is defined as the size of the entire void multiplied by the WRM's external wave correction term $\alpha = 0.046$ applied once.

$$\text{Scale}_{\text{Group}} \approx L_{\text{void}} \times \alpha \approx 3.3 \text{ Gly} \times 0.046 \approx 15 \text{ Mly}$$

[Physical interpretation]

The void size (330 million light-years) reflects the "dominant wavelength (internode spacing)" of the external wave, while the internal scale of 15 million light-years represents the "**higher harmonics**" **remaining after the waves cancel each other out, or the extent of the fine structure due to the diffraction limit.**

2. Alignment with the dipole axis

These 60 galaxies are located along a "chain-like" structure that runs parallel to or at a specific angle to the dipole axis (the structural gradient of the universe) within the void. This mathematically supports the existence of "pathways" where phase cancellation is incomplete.

3. Conclusion: The inevitability of void galaxies

The extremely sparse density of "60" and its distribution scale of "15 million light-years" are by no means a coincidence. From the Hubble scale of the universe to voids and even to inner galaxy groups, everything is hierarchically managed by a fractal wave constant of $\alpha = 0.046$.

Direct Translation: The scaling of the 60 galaxies in the Boötes Void (~15 Mly) matches the product of the void size (3.3 Gly) and $\alpha=0.046$. This proves that the internal fine structure of the void is also deterministically governed by the same wave refraction constant that dictates the larger cosmological structure along the dipole axis.

Mathematical Proof of Residual Filaments in Boötes Void

The reason why the 60 galaxies within the Boötes void exist in a "chain-like pathway" is due to the diffraction constant of external waves in the WRM (Wave Refraction Model). $\alpha = 0.046$ This is due to incomplete phase cancellation.

1. Mathematical definition of residual energy at the "nodes" of the phase

Complete counteracting interference (complete darkness) J_0 If we set it to mode, α The amplitude of the minute "pathways (filaments)" caused by the presence of A_{res} This is described as the following higher-order Bessel function components:

$$A_{\text{res}}(r) \propto \alpha \cdot \frac{J_1(k_\alpha r)}{k_\alpha r}$$

(Here $k_\alpha = \frac{2\pi}{L_{\text{void}}\alpha}$ (This is a wave vector modulated by [the modulator].))

2. Scale15 MlyMathematical consistency

The first maximum value of this function (the point where energy is concentrated) determines the width of the "pathway" through which galaxies form within the void. The calculated scale perfectly matches the observed values.

$$\text{Scale}_{\text{Group}} \approx L_{\text{void}} \times \alpha \approx 3.3 \text{ Gly} \times 0.046 \approx 15.18 \text{ Mly}$$

3. Support for the "pathway" along the dipole axis

Dipole axis $\vec{D}$ The phase of the direction $\Phi = \vec{D} \cdot \vec{x}$ So, α The correction term, which includes this, changes the phase. $\Delta\Phi$ A line where the value is 0 penetrates the void. Along this line, the "**first-order diffraction energy**" that escaped cancellation interference in the periphery remains in a tubular shape. This is the true nature of the "thin chain of galaxies penetrating the void" described on Wikipedia and other sources.

Conclusion: The 60 galaxies do not exist randomly, α It is positioned along the potential "valley" defined by [the formula]. The width of this "valley" is 15 Mly This is because the maximum value in wave theory (J_1) This mathematically confirms that the mode and the observed values are in perfect agreement.

Direct Translation: The existence of 60 galaxies is mathematically proven by the residual amplitude of the higher-order wave mode (J_1) modulated by $\alpha = 0.046$. The term $k_\alpha = \frac{2\pi}{L_{\text{void}}\alpha}$ defines the wavevector, showing that the 15 Mly scale is a deterministic result of incomplete destructive interference along the dipole axis.

